Cambridge (CIE) A Level Computer Science

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Programming Basics

Contents

- Pseudocode Basics

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Variables & Constants

What is a variable?

- A variable is an **identifier** that **can change** in the lifetime of a program
- Identifiers should be:
  - In **mixed case** (Pascal case)
  - Only **contain letters** (A-Z, a-z)
  - Only **contain digits** (0–9)
  - **Start with a capital** letter and not a digit
- A variable can be **associated a datatype** when it is declared
- When a variable is declared, **memory is allocated** based on the data type indicated

Pseudocode
DECLARE <identifier> : <datatype>

- To declare a variable, use the DECLARE keyword followed by the **name** and **data type**:

```
DECLARE Age : INTEGER
DECLARE Name : STRING
DECLARE IsLoggedIn : BOOLEAN
DECLARE Temperature : REAL
DECLARE DOB : DATE
```

- You can then **assign a value** using the assignment operator ← :

```
Age ← 18
Name ← "Alice"
IsLoggedIn ← TRUE
```

What is a constant?

- A constant is an **identifier set once in the lifetime of a program**
- Constants are generally named in **all uppercase characters**
- Constants **aid** the **readability** and **maintainability**

Pseudocode
CONSTANT <identifier> ← <value>

- To declare a constant, use the CONSTANT keyword:

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```
CONSTANT Pi ← 3.14159
CONSTANT MaxScore ← 100
CONSTANT SchoolName ← "Meridian Academy"
```

- Constants are **not reassigned** during execution – they are fixed values used throughout the algorithm

Example

- You are writing a program to calculate the area of a circle using the formula:

Area = π × radius²

- π is a **constant** (it never changes)
- radius and area are **variables** (they change depending on the input)

```
CONSTANT Pi = 3.14159

DECLARE Radius : REAL
DECLARE Area : REAL

OUTPUT "Enter the radius of the circle:"
INPUT Radius

Area ← Pi * Radius * Radius

OUTPUT "The area of the circle is: ", Area
```

Variables and constants in different languages

Feature	Python	VB.NET	Java
Declare variable	Just assign it (no keyword needed)	Dim radius As Double	double radius;
Assign variable	radius = 5	radius = 5	radius = 5;
Declare constant	Convention: use UPPERCASE (PI = 3.14) (not enforced)	Const Pi As Double = 3.14159	final double PI = 3.14159;
Reassign constant?	Yes (not truly constant unless enforced)	Cannot change once set	Cannot change once set
Use in calc.	area = PI * radius * radius	area = Pi * radius * radius	area = PI * radius * radius;

Arithmetic & logical operators

Arithmetic operators

- Arithmetic operators are used to perform basic **maths operations** in a program

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- These include adding, subtracting, multiplying and dividing values

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Common arithmetic operators

Operator	Purpose	Example	Result
+	Addition or string concatenation	5 + 3	8
		"John" + " " + "Doe"	"John Doe"
-	Subtraction	10 - 4	6
*	Multiplication	3 * 5	15
/	Division	10 / 2	5
^	Exponentiation (power of)	3 ^ 3	27
MOD	Modulus (remainder after division)	10 MOD 3	1

Operator precedence (BODMAS / BIDMAS)

- Arithmetic operators follow **operator precedence**
- Multiplication and division happen before addition and subtraction unless brackets are used

```
result ← 2 + 3 * 4        // gives 14
result ← (2 + 3) * 4      // gives 20
```

Logical operators

- Logical operators are used to **compare values**
- They return either **TRUE** or **FALSE** and are commonly used in conditions and loops

Common logical operators

Operator	Purpose	Example	Result
==	Equal to	5 == 6	FALSE
!=	Not equal to	5 != 7	TRUE
>	Greater than	5 > 10	FALSE
<	Less than	5 < 10	TRUE

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>=	Greater than or equal to	5 >= 10	FALSE
<=	Less than or equal to	5 <= 10	TRUE

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Example code

```
x ← 5
y ← 10

OUTPUT x == y      // FALSE
OUTPUT x != y      // TRUE
OUTPUT x < y       // TRUE
OUTPUT x > y       // FALSE
OUTPUT x <= y      // TRUE
OUTPUT x >= y      // FALSE
```


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